MOHAN KRISHNA KOLA

Software Engineer | Generative AI & ML Systems | Agentic AI, RAG & LLM Evaluation on AWS

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years designing, building, and operating production Generative AI systems and LLM platforms in Java and Python across regulated banking, e-commerce, and HIPAA healthcare. Builds multi-agent systems, RAG, embedding retrieval and ranking, LLM evaluation, guardrails, and MCP integrations on Amazon Bedrock, plus ML ranking models with online serving and A/B experimentation, balancing scalability, latency, security, and cost. Owns services end to end: design, implementation, evaluation-gated releases, monitoring, on-call, and root-cause fixes.

TECHNICAL SKILLS

GenAI & Agents: Multi-Agent Systems, Agentic Workflows, RAG, Model Context Protocol (MCP), Tool Calling, Prompt Engineering, LLMs (Claude, GPT), LangChain, LangGraph, Hugging Face

Retrieval & Evaluation: Embeddings, Hybrid Search & Reranking, Document Ingestion & Chunking, pgvector, Pinecone, FAISS, LLM Evaluation Test Sets, Tracing (LangSmith, Langfuse)

Machine Learning: ML Pipelines, Ranking & Recommendation Models, Online Model Serving, PyTorch, scikit-learn, pandas, MLflow, A/B Testing

AWS & Cloud: Amazon Bedrock (Runtime, Guardrails), Lambda, S3, CloudWatch, AWS Budgets, EC2, ECS, RDS, Multi-Region Architecture, Azure OpenAI, Azure AI Foundry, Microsoft Entra ID

Languages & Backend: Java, Python, C#, TypeScript, SQL, Spring Boot, Spring Security, FastAPI, REST, gRPC, Apache Kafka, Microservices, Object-Oriented Design, Design Patterns

Data, Security & Ops: PostgreSQL, MongoDB, Redis, RBAC, HIPAA/PHI, Audit Logging, Datadog, Docker, Kubernetes, CI/CD, JUnit

PROFESSIONAL EXPERIENCE

U.S. Bank | AI Engineer | Irving, TX **Jan 2026 – Present**

- Architected and built a governed GenAI Inference Gateway on Amazon Bedrock serving approved models to bank apps across two AWS regions.
- Built request- and response-interceptor AWS Lambda functions around every Bedrock Runtime call to inspect, govern, and audit-log model traffic, giving security and risk teams full traceability.
- Designed hybrid-cloud networking that routes model traffic across Azure OpenAI deployments, on-premises workloads, and multiple AWS accounts, with Microsoft Entra ID for identity and access control.
- Defined the Responsible AI guardrail program (Bedrock Guardrails configuration, approval process, go-live rules) and per-application approved-model access; took designs through architecture, security, and risk reviews.
- Operate the platform: CloudWatch and Datadog alarms, spend reporting, AWS Budgets cost controls, and app-team onboarding.

Wells Fargo | Software Engineer, Agentic AI **Jul 2025 – Dec 2025**

- Designed and built the multi-agent layer of an AI operations platform that ingests ServiceNow tickets and surfaces logs, alerts, runbooks, and LLM summaries, cutting median resolution time from 42 to 23 minutes (45%) across 500+ monthly incidents.
- Built MCP servers exposing 10+ internal tools (log search, alerts, runbooks, ticket updates) via tool calling, with human approval on every action.
- Designed the evaluation framework: a 200-case regression test set with traced agent runs that raised answer accuracy from 71% to 89% and gates every release on quality.
- Built a RAG assistant over 10K+ documents for 2,000+ employees, owning ingestion and chunking, hybrid keyword and vector retrieval with reranking, permission-aware filtering, and source citations; delivered it in Microsoft Teams through REST APIs.

Walmart | Full Stack Engineer (ML & Search) **Jan 2025 – Jun 2025**

- Built Support Assist, an NLP retrieval assistant for product specs, return windows, warranties, and runbooks, cutting average handle time from 8.5 to 6 minutes (29%) across 15K+ monthly customer issues.
- Built ML pipelines to train product ranking models (scikit-learn, MLflow), served them online via FastAPI, and designed the A/B framework (splits, success metrics, significance checks) that measured an 8% add-to-cart lift.
- Cut p95 latency by 220 ms and page load time 35% across search, catalog, and checkout via Redis caching and query tuning.

Wipro Limited (Client: Optum, UnitedHealth Group) | Software Engineer, Backend **Sep 2021 – Jul 2023**

- Developed 12+ Java 11/Spring Boot microservices for a HIPAA-compliant platform processing member, claims, and provider data at multi-million-record scale, securing PHI with Spring Security RBAC; built Claims & Coverage Assist, deflecting about 1,200 manual tickets per month.
- Designed REST, gRPC, and Kafka event-driven services; query tuning, Redis caching, and connection-pool optimization raised peak claims throughput from 1.2K to 1.7K requests/sec (42%).
- Automated Jenkins pipelines (release cycle 4 to 3 weeks), kept 80%+ JUnit coverage, ran code reviews, and owned on-call root-cause fixes.

PROJECTS

YuktiHire – GenAI Job Application Platform | TypeScript, React, Python, FastAPI, Claude/OpenAI APIs, PostgreSQL **yuktihire.com**

- Launched a Chrome extension and web platform that parses job postings and generates tailored resumes with an LLM, cutting per-application time from about 25 to under 5 minutes; built a provider-agnostic LLM layer over the Claude and OpenAI APIs with output validation.
- Built an ATS scoring engine that extracts required skills from each posting, scores the resume match, and ranks missing keywords; designed a Manifest V3 architecture with per-portal adapters so new job sites plug in without core changes.

EDUCATION

Master of Science, Mathematics and Statistics | GPA 4.0/4.0 | University of Texas at Arlington, TX **Aug 2023 – Dec 2024**

Bachelor of Engineering, Electrical and Electronics Engineering | Saveetha School of Engineering, Chennai, India **2017 – 2021**